

Initial Assessments of Covariate-Adjusted Average Treatment Effect Estimation under Misspecified Multiple Imputation Models

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- Introduction and motivation
- Model and simulation specifications
- Preliminary results
- Future directions

Causal Inference Review

Let $i \in \{1, \dots, n\}$ be an experimental unit.

- $T_i \in \{0, 1\}$: treatment status (control, treated)
- $Y_i(0)$: potential outcome under control
- $Y_i(1)$: potential outcome under treatment

The observed outcome is given by

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0).$$

The estimand is the **average treatment effect (ATE)**:

$$\tau = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0)).$$

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Note: $\hat{\tau}^{\text{adj}}$ frequently taken to be OLS estimate for T_i coefficient.

Setting: An analyst want to make inference about the effect of a treatment in a completely randomized experiment and has:

- Observed outcomes
- Treatment assignments
- *Pre-treatment* covariates

Randomization is “good” on average, but **regression adjustment** can help.

- Combat covariate imbalance [Ding, 2024]
- Improve estimator efficiency

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Suppose a binary covariate X_1 exhibits MCAR.

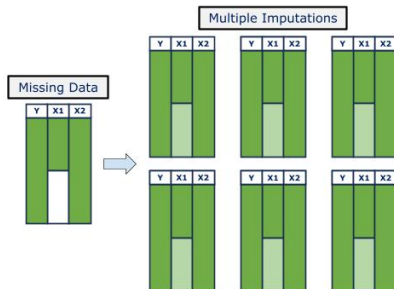
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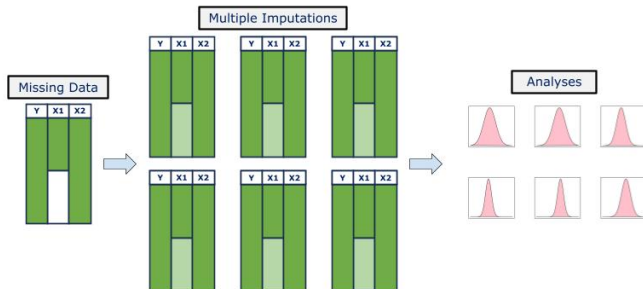
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Goal: Overcome missingness with **multiple imputation (MI)** to perform regression adjustment.

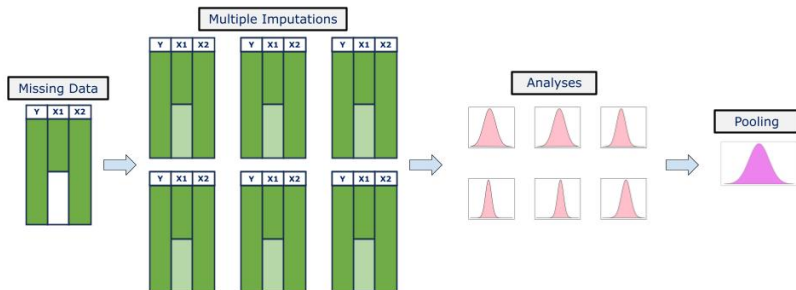
Multiple Imputation



Multiple Imputation



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Multiple Imputation

Why multiply impute?

- Use complete data methods easily
- Quantify imputation uncertainty
- Excellent for sensitivity analysis

For $n = 10,000$ and $n_1 = 5,000$, we generate data as follows.

Treatment and Covariates:

$$\mathbb{P}(\mathbf{T} = \mathbf{t}) = \frac{1}{\binom{n}{n_1}} \quad t_i \in \{0, 1\}, \quad \|\mathbf{t}\|_1 = n_1$$

$$X_{1i} \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(0.5)$$

$$X_{2i} \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(0.5)$$

Outcomes:

$$Y_i(0) \mid X_{1i}, X_{2i} \stackrel{\text{ind.}}{\sim} \mathcal{N}(\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_{12} X_{1i} X_{2i}, \sigma^2)$$

$$Y_i(1) = Y_i(0) + \tau$$

$$\Downarrow$$

$$Y_i \mid T_i, X_{1i}, X_{2i} \stackrel{\text{ind.}}{\sim} \mathcal{N}(\beta_0 + \tau T_i + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_{12} X_{1i} X_{2i}, \sigma^2)$$

Missingness: Let $R_i = 1$ for observed X_{1i} and $R_i = 0$ for missing X_{1i} :

$$R_i \mid X_{1i} \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(0.3)$$

Data-Generating Model

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Sensitivity analysis:

$$\tau, \beta_1, \beta_2 \in \{0.5, 7.5\} \quad (\text{"low" and "high" values})$$

$$\beta_{12} \in \{0, 0.5, 1, \dots, 9.5, 10\} \quad (\text{interaction strength})$$

Imputation Mechanism

First iteration:

- 1 Draw $p_X \sim \text{Beta}(1, 1)$.
- 2 Impute missing $X_{1i} \mid p_X \sim \text{Bernoulli}(p_X)$.
- 3 Calculate OLS coefficient and variance estimates.

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Subsequent iterations:

- 1 Draw

$$p_X \sim \text{Beta} \left(1 + \sum_{R_i=0} \mathbb{I}[X_i = 1], 1 + \sum_{R_i=0} \mathbb{I}[X_i = 0] \right).$$

- 2 Impute missing X_{1i} based on p_X draw, imputation model, and previous OLS estimates.
- 3 Calculate new OLS coefficient and variance estimates.

Regression/MI Model Combinations

Recall: The regression model produces the ATE estimates, while the MI model completes the data set ahead of the regression.

Regression Model	MI Model	Imputation Contributors
Full	Full	$p_X, Y, T, X_1, X_2, X_1X_2$
	No interaction	p_X, Y, T, X_1, X_2
	X_1 only	p_X, Y, T, X_1
No interaction	No interaction	p_X, Y, T, X_1, X_2

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For each missing X_{1i} , calculate

$$\mathbb{P}(X_{1i} = 1 \mid \cdots) = \frac{w_1}{w_1 + w_0},$$

where w_1 and w_0 are based on p_X and the OLS estimates.

Simulation Set-up

- 1 Generate $n = 10,000$ experimental units.
- 2 Assign $n_1 = 5,000$ to treatment.
- 3 Induce missingness in X_1 at random ($p = 0.3$).
- 4 Generate outcomes for each combination of coefficients.
 - 1 Calculate unadjusted ATEs.
 - 2 Run imputation mechanism and calculate ATEs for all regression-MI models across several iterations.

Repeat entire process across several seeds and calculate MSEs.

High ATE Results: $\log(\text{MSE ratio})$

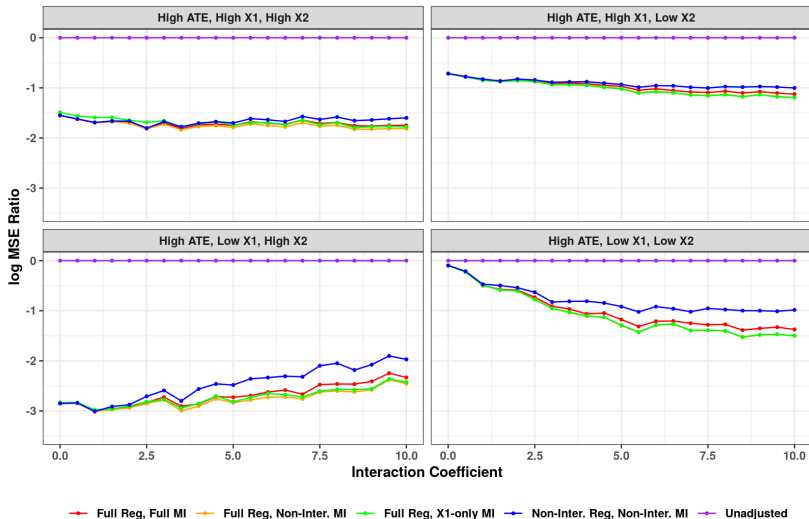


Figure: Log-transformed adjusted-to-unadjusted MSE ratios vs. interaction strength for all high-ATE models.

Low X_1 Coefficient Results: $\log(\text{MSE ratio})$

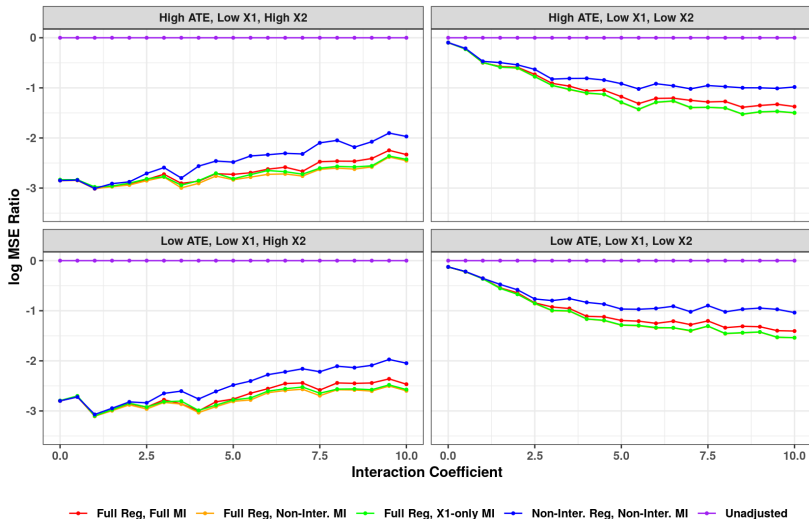


Figure: Log-transformed adjusted-to-unadjusted MSE ratios vs. interaction strength for all low- X_1 coefficient models.

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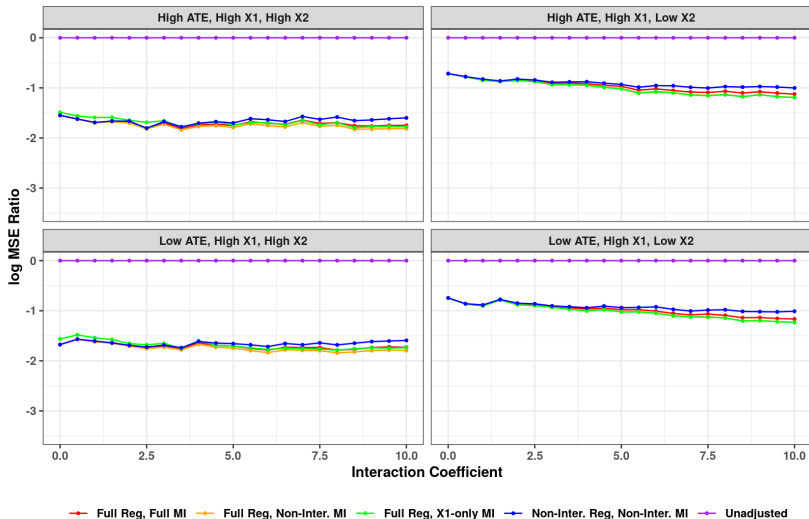


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Summary

- Unadjusted estimator consistently outperformed by all adjusted alternatives
 - Closest at low X_1 and X_2 coefficient values
- Interaction-inclusive MI model frequently outperformed by both interaction-exclusive MI models
- Greater variation in performance at lower-magnitude interaction strengths
- Need more simulations and further analysis for most grounded conclusions

Future Directions

- Examine broader range of coefficient values
- Consider alternative settings
 - Covariate dependence structure
 - Transformed data
- Compare to additional estimators
 - Centered covariates [Lin, 2013]
 - Missingness indicator method [Zhao and Ding, 2024], [Zhao et al., 2023]
- Identify diagnostics for guidance



Ding, P. (2024).

A First Course in Causal Inference.

CRC Press.



Lin, W. (2013).

Agnostic notes on regression adjustments to experimental data: reexamining freedman's critique.

The Annals of Applied Statistics, 7(1):295–318.



Zhao, A. and Ding, P. (2024).

To adjust or not to adjust? estimating the average treatment effect in randomized experiments with missing covariates.

Journal of the American Statistical Association,
119(545):450–460.



Zhao, A., Ding, P., and Li, F. (2023).

Covariate adjustment in randomized experiments with missing outcomes and covariates.

arXiv preprint arXiv:2311.10877v2.

Appendix: Imputation Probability Computations

Full MI:

$$w_1 = p_X \cdot \text{dnorm} \left(Y_i; \hat{\beta}_0 + \hat{\tau}^{\text{adj}} T_i + \hat{\beta}_1 + \left(\hat{\beta}_2 + \hat{\beta}_{12} \right) X_{2i}, \hat{\sigma}^2 \right)$$

$$w_0 = (1 - p_X) \cdot \text{dnorm} \left(Y_i; \hat{\beta}_0 + \hat{\tau}^{\text{adj}} T_i + \hat{\beta}_2 X_{2i}, \hat{\sigma}^2 \right)$$

No-interaction MI:

$$w_1 = p_X \cdot \text{dnorm} \left(Y_i; \hat{\beta}_0 + \hat{\tau}^{\text{adj}} T_i + \hat{\beta}_1 + \hat{\beta}_2 X_{2i}, \hat{\sigma}^2 \right)$$

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X_1 -only MI:

$$w_1 = p_X \cdot \text{dnorm} \left(Y_i; \hat{\beta}_0 + \hat{\tau}^{\text{adj}} T_i + \hat{\beta}_1, \hat{\sigma}^2 \right)$$

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Appendix: Low ATE Results; $\log(\text{MSE ratio})$

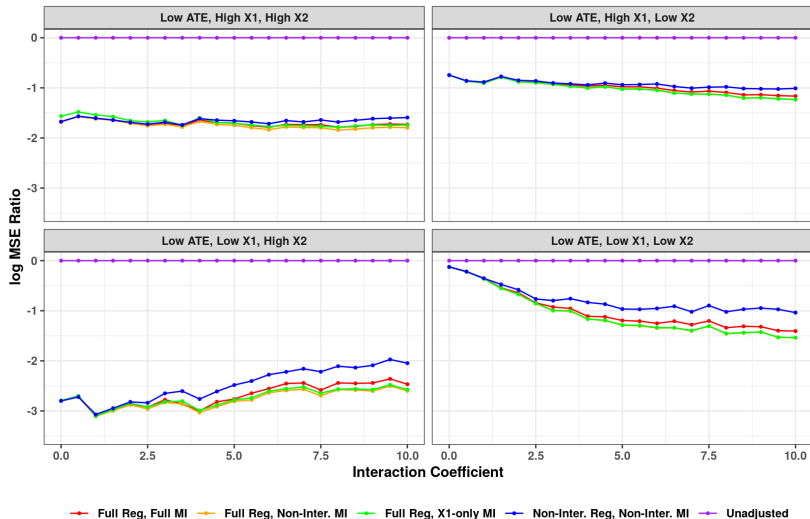


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Appendix: Low X_2 Coefficient Results; $\log(\text{MSE ratio})$

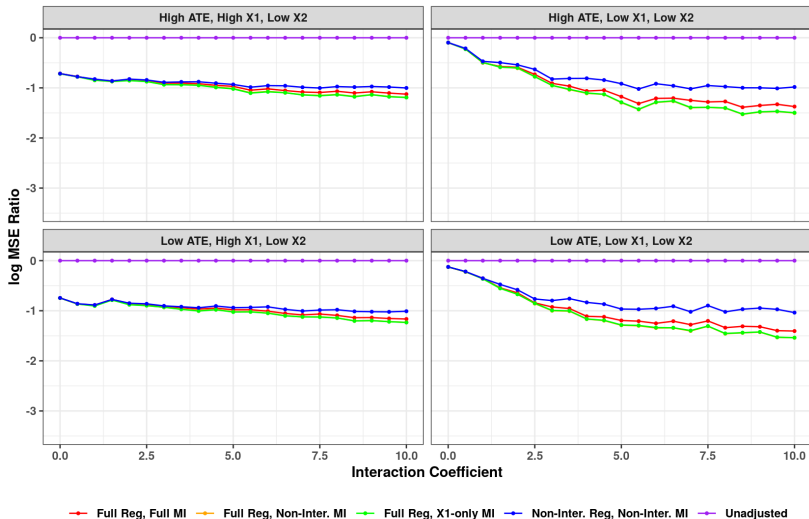


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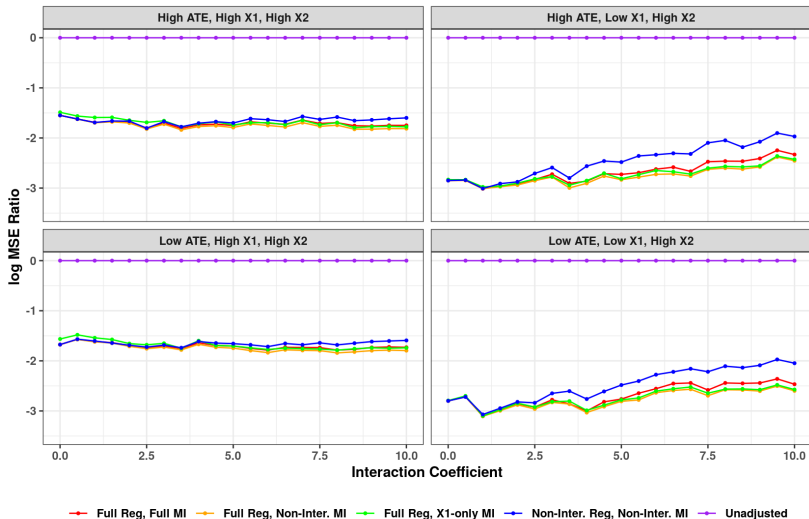


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